Harrison Lin

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Education

Georgia Institute of Technology | Atlanta, GA — 2025 – Present, Graduating May 2028

- Bachelor of Science in Electrical Engineering; GPA 3.8/4.0
- Concentration in Circuit Technology and Electronic Devices
- Related Coursework: Digital Design Lab, Signal Processing, Microelectronics, Circuit Analysis, Embedded Systems
- Activities and Societies: Theta Chi Fraternity, Propulsive Landers, Yellow Jacket Fencing Club, GT Navigators

Virginia Polytechnic Institute and State University | Blacksburg, VA — 2024 – 2025

- Bachelor of Science in Electrical Engineering; GPA 3.8/4.0
- Related Coursework: Digital System Design, Computational Engineering, Differential Equations, Physics

Skills

- **Hardware/Interfaces:** FPGA, PCB Design, I²C, UART, CAN, SPI, Digital Logic Design, Electrical Schematics, Oscilloscope, Soldering
- **Programming/HDL:** Verilog, VHDL, C++, Python, MATLAB, Java
- **Software:** Quartus Prime, KiCad, LTSpice, Siemens NX, Capital Harness Systems, Simulink/Stateflow, Tidy3D FDTD, Fusion360
- **Engineering:** Power Electronics, Electrical Load Analysis, Circuit Analysis, Grounding and Bonding, Hardware Troubleshooting

Related Engineering Experience

Electrical Engineering Intern | L3Harris Technologies | Greenville, TX — Jan 2026 – Jul 2026

- Engineered aircraft power distribution across 5+ aircraft configurations by analyzing electrical loads, wire sizing, and circuit protection
- Developed 50+ wiring diagrams including 75%+ of an aircraft baseline's documentation, specifying connectors, shielding, and grounding
- Resolved 90%+ of ground-station change orders by diagnosing wiring faults, pinout discrepancies, and electrical integration issues
- Authored reports evaluating ignition hazards, power loads, and electrical faults to support FAA airworthiness and safety compliance

Projects

FPGA UART Transceiver | Independent Personal Project| Verilog, Quartus, UART, FPGA

- Designed a synthesizable UART transceiver using FSMs, counters, and shift registers for 8-N-1 serial communication
- Implemented UART transmit and receive datapaths with asynchronous input synchronization, mid-bit sampling, framing/error detection
- Deployed the design to FPGA hardware to validate bidirectional PC communications through UART loopback interface

Custom MIDI DJ Controller | Independent Personal Project | KiCad, I²C, Arduino, C/C++

- Designed a custom 2-deck MIDI DJ controller PCB in KiCad, integrating 20 analog controls, 16 performance pads, and LED feedback
- Architected mixed-signal I/O using I²C GPIO expanders and analog multiplexers, incorporated power decoupling and pull-up networks
- Developed Arduino USB-MIDI firmware to scan multiplexed inputs, debounce controls, and interface with RekordBox DJ Software

Additional Engineering Experience

PCM Subteam Member | Virginia Tech Hybrid Electric Vehicle Team | Blacksburg, VA — Jan 2025 – May 2025

- Developed a closed-loop PID adaptive cruise controller in MATLAB/Simulink to regulate vehicle speed and minimize tracking error
- Integrated CAN-based signal transmission and reception between vehicle control subsystems using MATLAB/Simulink
- Performed model-in-the-loop testing and software coverage analysis to validate propulsion control functionality and safety

Research Intern | University of Maryland, College Park | College Park, MD — Mar 2024 – Aug 2024

- Modeled nanophotonic waveguides, optical cavities, and photon emitters using finite-difference time-domain simulations
- Analyzed optical reflection, transmission, and electromagnetic field distributions to characterize photonic device behavior
- Presented simulation results and photon projection calculations to graduate researchers investigating quantum photonic devices

Leadership

Alumni Relations Chair | Theta Chi Fraternity, Alpha Nu Chapter | Atlanta, GA — May 2026 – Present

- Led alumni outreach and engagement, connecting 100+ active members and alumni through events and communications.
- Managed alumni communications through LinkedIn and direct outreach, increasing alumni involvement in career development by ~2x.